

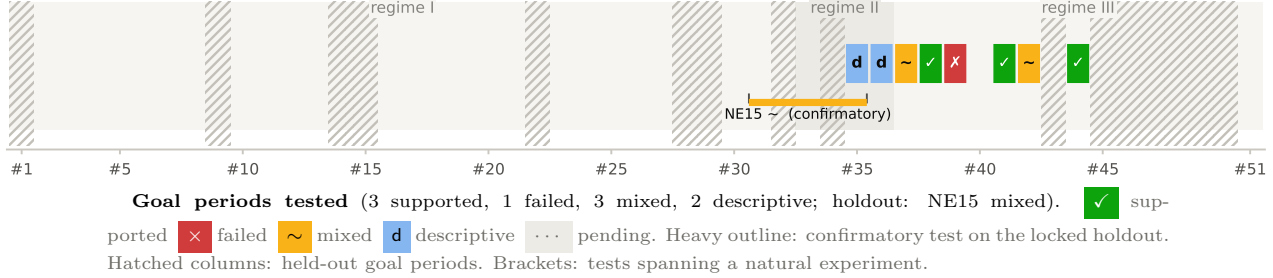
H05: Cutting the cross-room channel lowers entropy production, and rooms become coupled blocks

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 60% Faithfulness 2.5/5 Usefulness 2.5/5

Bottom line. Rooms cut the chat coupling, as the scaffold implies, detectably in a partly blind test; but work is not room-structured, entropy production is unmeasurable at 1-min resolution, and the confirmatory verdict was inconclusive.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

Do rooms make agents coupled blocks? Round 1 (exploratory, 99 non-holdout days, #35–#44) fitted 1-min talk and activity spins. Within-room talk coupling beat cross-room in 6/6 regime-III two-room windows (kinetic-Ising J , Fisher $p < 10^{-4}$), but per period the primary κ_x test passed only in G38, G41, G44 (mixed G37, G42; failed G39). The pooled two-way-FE effect of co-location was $\beta \approx +0.008$ ($z = 2.6$). Activity spins: not robust. Entropy production stayed at the cross-day null in 15/18 windows, so it cannot be measured at this resolution. Holdout, NE15 (03-16 #best/#rest split): the primary pair DiD passed (talk $\kappa_x -0.022 [-0.038, -0.008]$, $p_{\text{perm}} = 0.02$) and J_{out} fell to ≈ 0 . The pooled TWFE missed ($z = 1.4$), so the pre-registered verdict is **inconclusive**. Unpredicted: within-room J_{in} rose about $\times 6$. Caveat: the post-split window (#35) was not held out and had already been seen in round 1.

$$P(s'_i | \mathbf{s}) = \frac{e^{s'_i H_i}}{2 \cosh H_i}, \quad H_i = h_i + \sum_j J_{ij} s_j$$

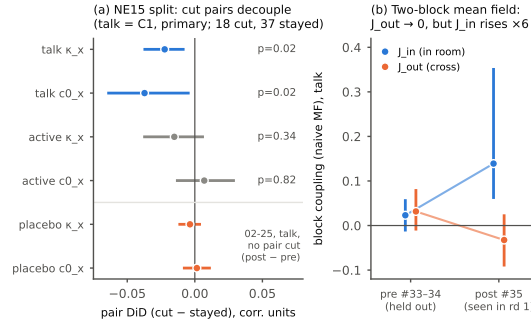
$$J_{ij} = J_{\text{in}} \delta_{r_i r_j} + J_{\text{out}} (1 - \delta_{r_i r_j})$$

$$\sigma = \sum_{i < j} (J_{ij} - J_{ji}) \langle g_{ij} \rangle, \quad g_{ij} = s'_i s_j - s_i s'_j$$

$$\Sigma_g = \max_{\theta} [\theta \cdot \langle g \rangle - \ln \langle e^{-\theta \cdot g} \rangle] \leq \sigma$$

$$J^{\text{MF}} = D^{-1/2} (I - \hat{R}^{-1}) D^{-1/2}$$

$$\kappa_{ij,d} = \alpha_{ij} + \gamma_d + \beta \text{coloc}_{ij,d}$$



Observables. (a) Confirmatory pair difference-in-differences at the 03-16 split: the coupling change of the 18 pairs put in different rooms minus that of the 37 pairs that stayed together (pre #33–34, held out; post #35, not held out). Talk coupling dropped (C1 passed); activity did not. The 02-25 placebo, which separated no one, shows no change. Bars: day-bootstrap 95% CIs; p : assignment permutation. (b) Block mean field: cross-room J_{out} falls to ≈ 0 , while within-room J_{in} rises (unpredicted).

PHYSICAL MODEL

Kinetic Ising (model 02). $s_i(t) = \pm 1$: agent i talked (talk spin) or acted (activity spin) in 1-min bin t ; $s'_i = s_i(t+1)$. h_i : daily-schedule field (removed by a cross-day surrogate). r_i : room; J_{in} , J_{out} : same- and cross-room coupling; a cut predicts $J_{\text{out}} \rightarrow 0$. Σ_g : Aguilera–Ito–Kolchinsky lower bound on the entropy production σ , dual weights θ . $\hat{R}$, D : block correlations, variances. α_{ij} , γ_d : pair, day effects.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Candidly, agents who can't see each other's chat stop timing their chat together: close to a restatement of the scaffold, since rooms are a visibility filter. Three things are mildly useful. (1) A cheap monitor: a pair DiD on 1-min talk spins caught the cut with 18 cut pairs and 5 post-split days (power only about 25–60%). (2) A negative result an operator splitting teams should know: activity, the actual work, is *not* room-structured. Rooms reorganize conversation, not labor. (3) The unpredicted $\times 6$ rise in within-room coupling hints that attention is conserved and redirected to roommates (HH90, now tested by H18). The thermodynamic half, entropy production, is unmeasurable here and buys nothing practical.